

MATHEMATICS – CLASS X

2013

1. Answer all the questions

(1X6=6)

- (i) If the ratio of cost price and selling price of an article is 5 : 4, what is the percentage of loss?
- (ii) $(1 - x)^2 + y^2 = 0$, what is the value of $(x + y)$? (x, y are real numbers.)
- (iii) $(x^2 - 20x - a) = 0$ will be a perfect square when the value of 'a' will be:
(a) 100 (b) -100 (c) 0 (d) 80
- (iv) If the measure of an angle of a cyclic quadrilateral is 75° , what is the measure of angle opposite to the given angle?
- (v) In which quadrant does the point (4, -3) lie?
- (vi) What is the value of $(\cos^2 20^\circ + \cos^2 70^\circ)$?

2. Answer all the questions.

(2X7=14)

- (i) Average age of 7 boys is 7 years. When another boy joins them, then the average of these 8 boys becomes 8 years. What is the age of the new boy?
- (ii) If 75% of A = 40% of B, what is the value of (A : B)?
- (iii) If $x + \frac{9}{x} = 6$, determine the numerical value of x^2 .
- (iv) Determine the greatest value of x from the following inequation: $2x - 2 \geq 3x + 5$.
- (v) ABC is an equilateral triangle. Side BC is produced to D such that CD = AB. What is the measure of $\angle BAD$?
- (vi) AB is a diameter of a cyclic quadrilateral ABCD. If $\angle ACD = 50^\circ$, determine the measure of $\angle BAD$.
- (vii) If $\sin 3x = 1$, what is the value of $\tan 2x$?

3. Answer any TWO.

(5X2=10)

- (a) A mixture of dettol water contains 80% water. What volume of dettol is to be added to 60 litres of this mixture, so that the new mixture may contain 75% water?
- (b) The price of coal is increased by 25%. If a family wants to keep the same monthly expenditure for coal, what percentage of coal consumption is to be reduced?
- (c) Bimalbabu lent some amount of money for 4 years at 8% simple interest per annum to a person and lent double the money for the same period at 10% simple interest per annum to another person. Bimalbabu got Rs 960 more as interest from the second person than the first person. What is the total amount of money lent by Bimalbabu?
- (d) The compound interest on Rs. 5000 for two years is Rs. 408. Find the rate of interest per annum.

4. Answer any ONE question:

(4X1=1)

- (a) Find LCM of: $3a^2 - 5ab - 12b^2$; $a^5 - 27a^2b^3$; $9a^2 + 24ab + 16b^2$.
- (b) Find HCF of: $x^3 - 4x$; $4x^2 - 20x + 24$; $x^2 - 4x + 4$.

5. Solve any ONE

(3X1=3)

- (a) $x - 2y = 24$; $x - 8y = 48$. [Using the method of elimination or of cross multiplication.]
- (b) $\frac{1}{x} - \frac{1}{3} = \frac{1}{x+2} - \frac{1}{5}$.

6. Answer any ONE question:

(4X1=4)

- (a) Plants are planted in rows in a garden in such a way that the total number of rows exceeds the number of the plants in each row by 5. If the total number of plants in the garden is 336, find the number of the plants in each row.
- (b) If the product of two consecutive odd numbers is 783, determine the two numbers.

7. Draw the graphs of the inequalities and indicate the solution region (any ONE). (4X1=4)

- (a) $x \geq 0$; $y \geq 0$ and $2x + 3y \leq 12$.
- (b) $x + y \leq 5$ and $x + y \geq -5$.

8. Answer any ONE question:

(3X1=3)

(a) If $a : b = b : c$, show that $(a + b)^2 : (b + c)^2 = a : c$.

(b) If $a^2by + cz$, $b^2 = cz + ax$, $c^2 = ax + by$, show that $\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z} = 1$.

9. Answer any ONE question:

(3X1=3)

(a) y varies directly with squares of x , and $y = 9$ when $x = 9$. What is the value of x when $y = 4$?

(b) If $(a + b) \propto \sqrt{ab}$, show that: $(\sqrt{a} + \sqrt{b}) \propto (\sqrt{a} - \sqrt{b})$.

10. Answer any ONE question:

(3X1=3)

(a) If $x = 3 + \sqrt{3}$ and $xy = 6$, find the value of $(x + y)^2$.

(b) Find the simplest value of: $\frac{\sqrt{2}(2 + \sqrt{3})}{\sqrt{3}(\sqrt{3} + 1)} - \frac{\sqrt{2}(2 - \sqrt{3})}{\sqrt{3}(\sqrt{3} - 1)}$.

11. Answer any TWO questions:

(5X2=10)

(a) Prove that the angle which an arc of a circle subtends at the centre, is double the angle subtended by it at any point on the circle.

(b) Prove that the area of the squares on the hypotenuse of a right triangle is equals to the sum of the areas of the squares on the other sides of the triangle.

(c) Prove that the tangent to a circle at any point on it is perpendicular to the radius through the point of contact.

12. Answer any ONE question:

(3X1=3)

(a) There are two concentric circles such that two chords AB and AC of greater circle touch the smaller at P and Q respectively. Prove that $PQ = \frac{1}{2} BC$.

(b) PQR is a triangle and X is a mid point of PQ. If a line through X parallel to QR, intersect line PR at Y, prove that Y is the mid point of PR.

13. Answer any ONE question:

(5X1=5)

- (a) Draw a line segment AB of length 4 cm. Draw two circles of radius 2 cm each with centres A and B. Also draw any one direct common tangent to the circles.
- (b) Draw an equilateral triangle having each side of lengths 7 cm. Also draw its incircle.

14. Answer any ONE question:

(3X1=3)

- (a) The base of a right prism is a triangle having sides of lengths 12 cm, 16 cm and 20 cm. The volume of the prism is 1152 cubic cm. Find the height of the prism.
- (b) The area of the curved surface of a right circular cone is $\sqrt{10}$ times the area of its base. Show that the height of the cone is three times the radius of the base.

15. Answer any ONE question:

(4X1=3)

- (a) The area of curved surface of a solid right circular cylinder is 1320 sq cm. If the diameter of the base of the cylinder is 14 cm, find its height.
- (b) The base of the right pyramid is a rectangle whose length and breadth are respectively 12 m and 9 m. If the length of the slant height of the pyramid is 8.5 m, find the volume of the pyramid.

16. Answer any TWO questions:

(3X2=6)

- (a) Find the circular measures of $52^{\circ}52'30''$.
- (b) Find the value of $\tan \theta$ from the relation $5\sin^2 \theta + 4\cos^2 \theta = \frac{9}{2}$ where $0^{\circ} < \theta < 90^{\circ}$.
- (c) If $x = \sin^2 30^{\circ} + 4\cot^2 45^{\circ} - \sec^2 60^{\circ}$, find the value of x .
- (d) If $\tan \alpha = \cot \beta$, find the value of $\cos(\alpha + \beta)$, where $0^{\circ} < \alpha, \beta < 90^{\circ}$

17. Answer any ONE question:

(5X1=5)

- (a) From a point on the roof of a house, it is observed that the angle of depression of the top and the foot of a lamp post are respectively 30° and 60° . Find the ratio of the height of the house and that of the lamp post.
- (b) The ratio of the heights of two towers 1:3. The angle of the top of the bigger tower from the foot of the smaller tower is 60° . Find the angle of elevation of the top of the smaller tower from the foot of the bigger tower.
